

● Team project (**Reconfigurable FIR filter**)

■ Project plan

◆ 제출 기한

- ~11/25 KST 23:59

◆ 포함 내용

- Project 진행 계획
- 설계 구조 초안
- 팀 구성원들의 역할 분담

◆ 제출 방법

- 개인 e-mail: inchul.song@ssu.ac.kr

■ Project report

◆ 제출 기한

- 팀별 발표 순선 fix 예정 (TBD.)
- 팀별 일정에 따라 14주차(~12/02, KST 23:59) 또는 15주차(~12/09, KST 23:59)

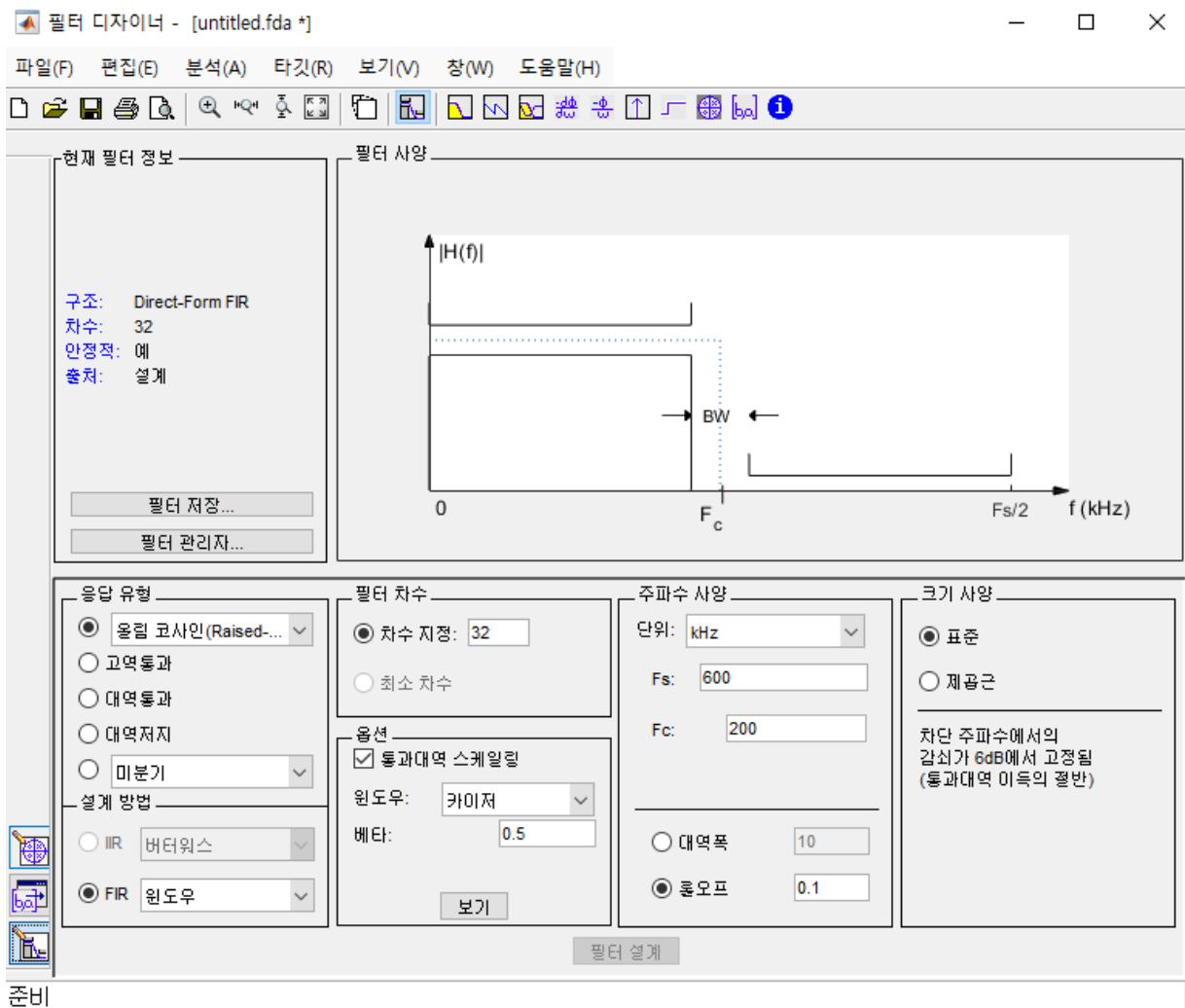
◆ 포함 내용

- Project 진행 결과
- 최종 설계 구조
- 팀 구성원들의 역할 분담
- Verilog/Testbench/waveform
- 구조 개선안 및 소감 등

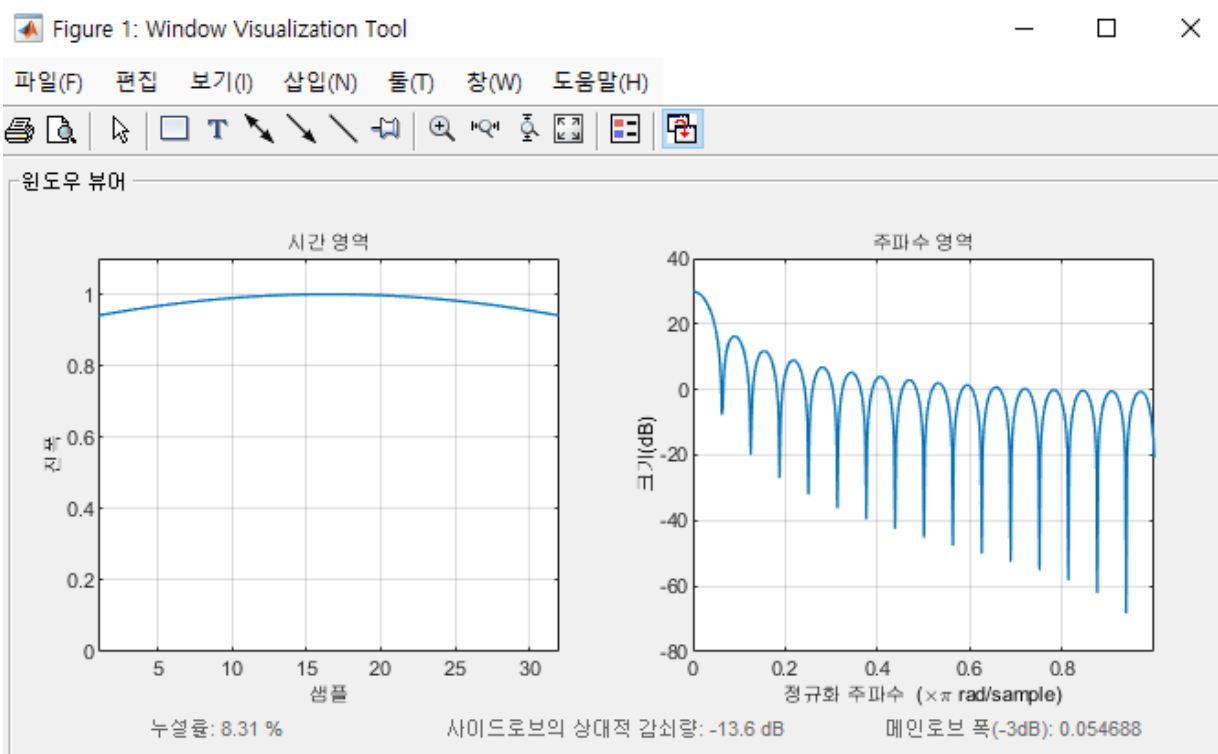
1. FIR Filter algorithm design (개인 Project와 동일)

1) FIR filter 사양

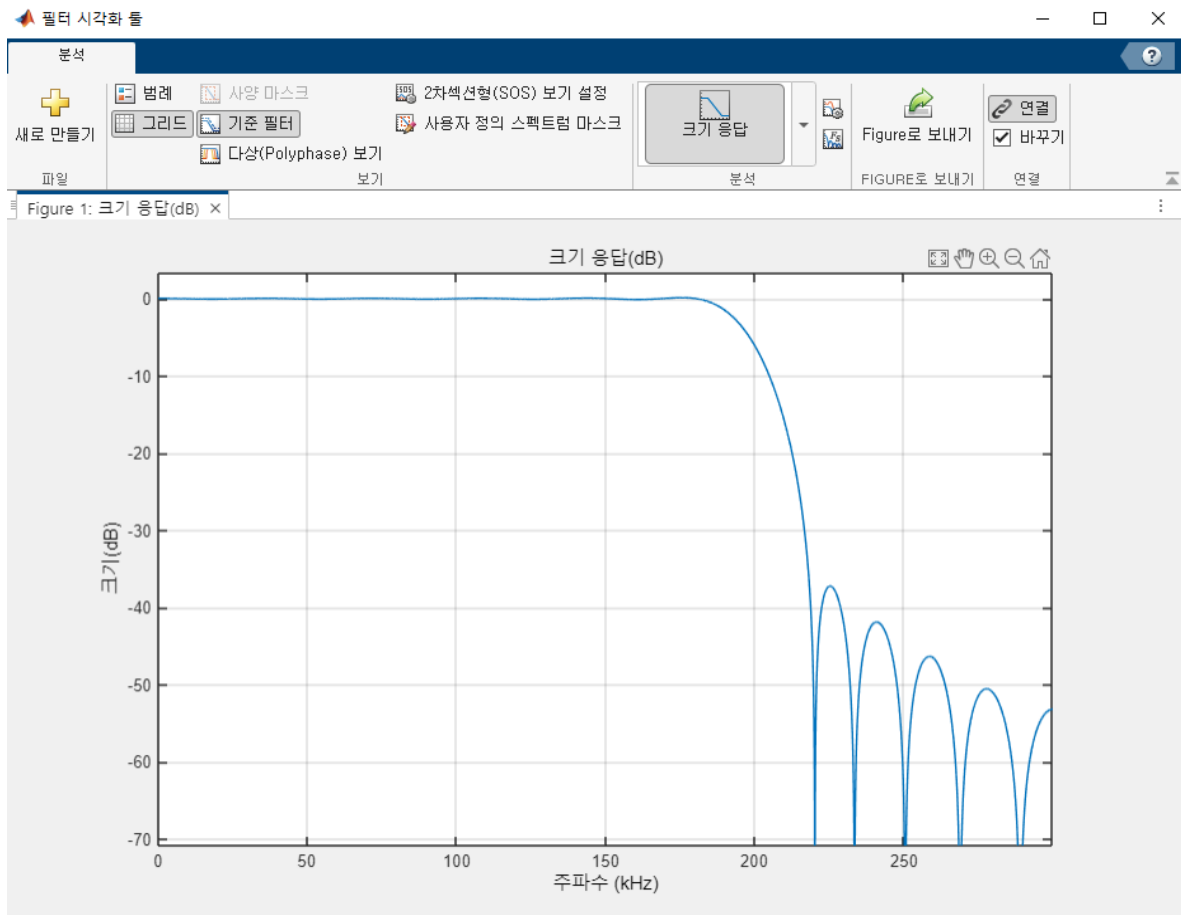
- ➔ Sampling rate: 600kHz
- ➔ Symbol rate: 200kHz
- ➔ Symbol: -3(3'b101), -1(3'b111), +1(001), +3(011) <- signed



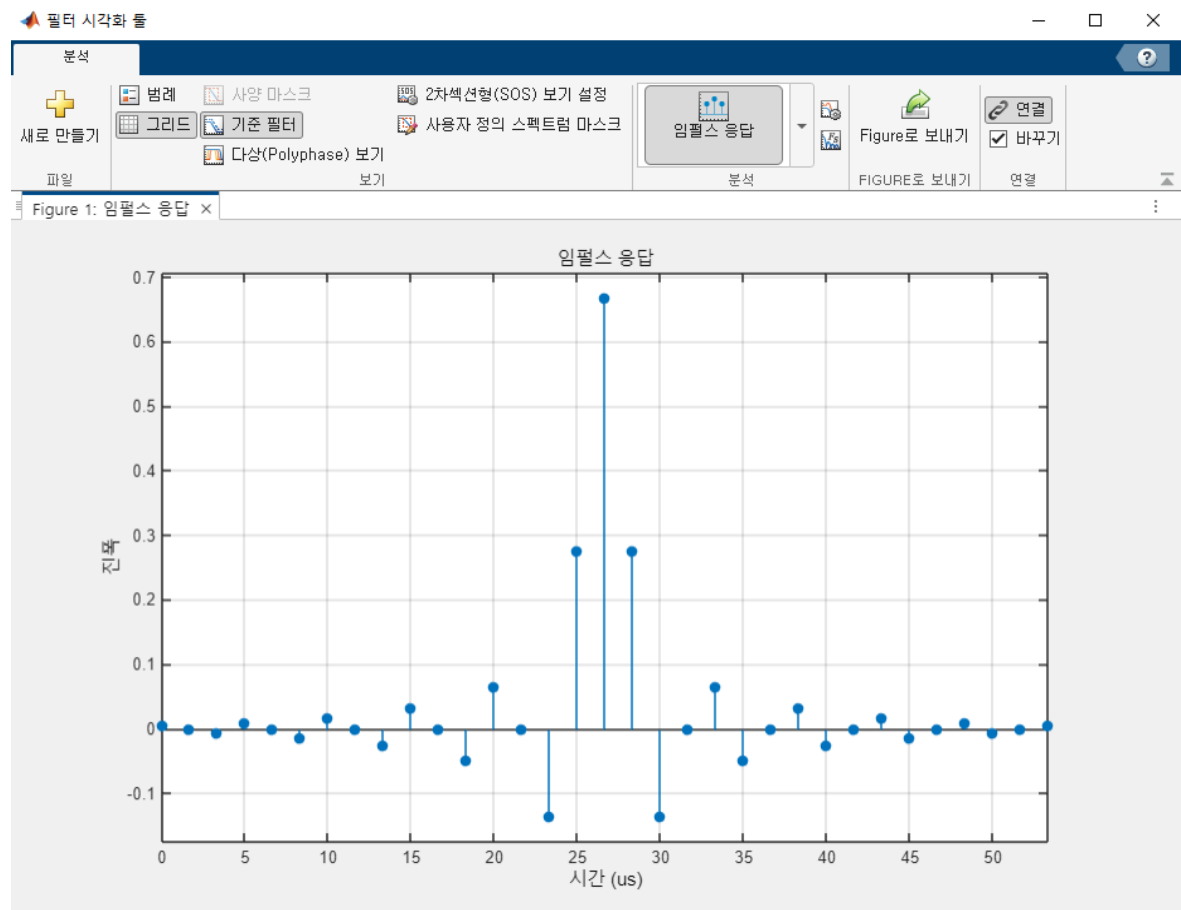
2) Kaiser window (<https://kr.mathworks.com/help/signal/ug/kaiser-window.html>)



3) Frequency response



4) Impulse response



5) Filter coefficient

분자:

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0.004462417558193448700876171386653368245
-0.000000000000000008207265074504004771501
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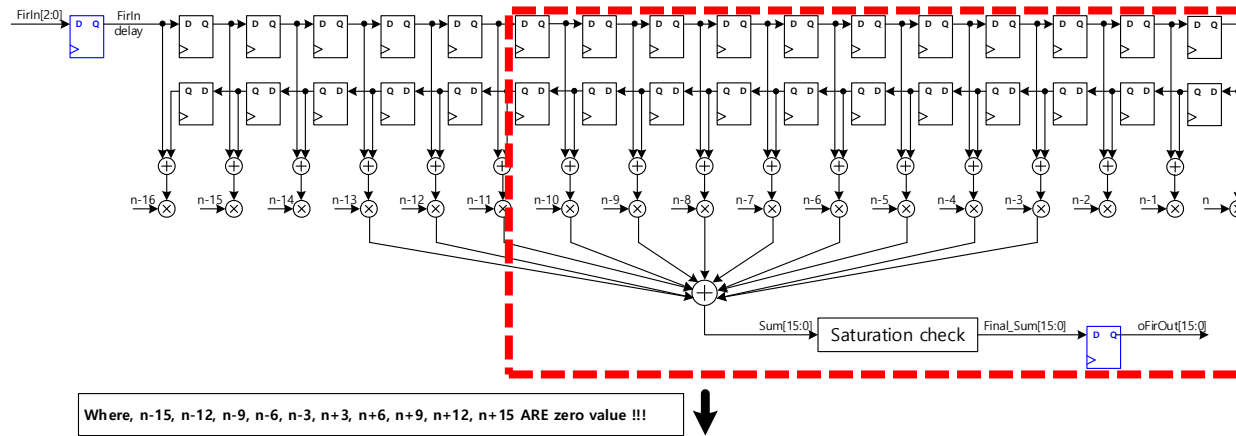
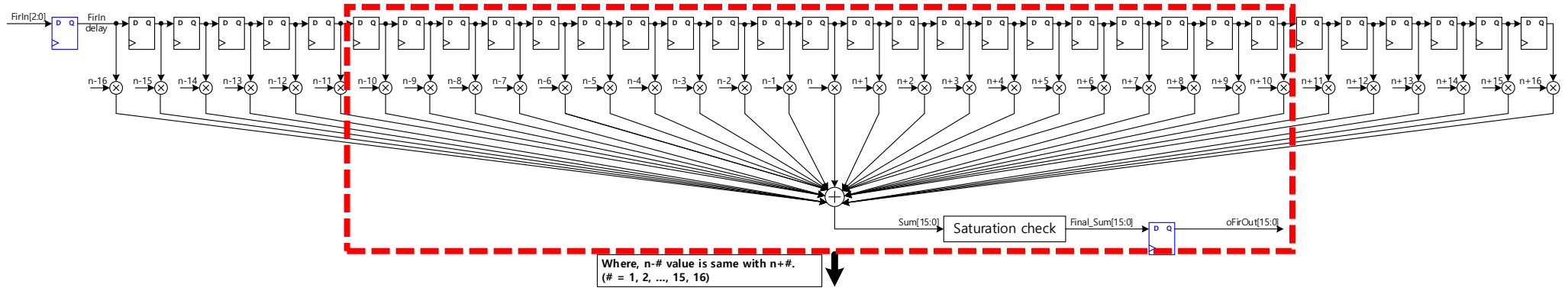
→ Normalized filter tap (33ea -> 21ea)

Tap Num.	소수 자릿수 : 20	소수 자릿수 : 10	부호	절대값	절대값*750	Integer	Dec2Bin	16bit tap coefficient[15:0]	16bit tap coefficient[8:0]	Etc.
n-16	0.00446241755819344000	0.0044624176	양	0.0044624176	3.3468131686	3	11	0000_0000_0000_0011	0_0000_0011	
n-15	-0.00000000000000000821	0.0000000000	0	0.0000000000	0.0000000000	(0)	0	0000_0000_0000_0000	0_0000_0000	Symbol
n-14	-0.00739666096745547000	-0.0073966610	음	0.0073966610	5.5474957256	6	101	0000_0000_0000_0101	0_0000_0101	
n-13	0.00928151419730391000	0.0092815142	양	0.0092815142	6.9611356480	7	110	0000_0000_0000_0110	0_0000_0110	
n-12	-0.00000000000000001302	0.0000000000	0	0.0000000000	0.0000000000	(0)	0	0000_0000_0000_0000	0_0000_0000	Symbol
n-11	-0.01415300566769760000	-0.0141530057	음	0.0141530057	10.6147542508	11	1010	0000_0000_0000_1010	0_0000_1010	
n-10	0.01730414582010220000	0.0173041458	양	0.0173041458	12.9781093651	13	1100	0000_0000_0000_1100	0_0000_1100	
n-9	-0.00000000000000001790	0.0000000000	0	0.0000000000	0.0000000000	(0)	0	0000_0000_0000_0000	0_0000_0000	Symbol
n-8	-0.02574775098543550000	-0.0257477510	음	0.0257477510	19.3108132391	19	10011	0000_0000_0001_0011	0_0001_0011	
n-7	0.03156805776840270000	0.0315680578	양	0.0315680578	23.6760433263	24	10111	0000_0000_0001_0111	0_0001_0111	
n-6	-0.00000000000000002212	0.0000000000	0	0.0000000000	0.0000000000	(0)	0	0000_0000_0000_0000	0_0000_0000	Symbol
n-5	-0.04932627602752730000	-0.0493262760	음	0.0493262760	36.9947070206	37	100100	0000_0000_0010_0100	0_0010_0100	
n-4	0.06420091371232200000	0.0642009137	양	0.0642009137	48.1506852842	48	110000	0000_0000_0011_0000	0_0011_0000	
n-3	-0.00000000000000002498	0.0000000000	0	0.0000000000	0.0000000000	(0)	0	0000_0000_0000_0000	0_0000_0000	Symbol
n-2	-0.13542712872673600000	-0.1354271287	음	0.1354271287	101.5703465451	102	1100101	0000_0000_0110_0101	0_0110_0101	
n-1	0.27445588248551800000	0.2744558825	양	0.2744558825	205.8419118641	206	11001101	0000_0000_1100_1101	0_1100_1101	
n	0.66666666666666660000	0.6666666667	양	0.6666666667	500.0000000000	500	111110011	0000_0001_1111_0011	1_1111_0011	Symbol
n+1	0.27445588248551800000	0.2744558825	양	0.2744558825	205.8419118641	206	11001101	0000_0000_1100_1101	0_1100_1101	
n+2	-0.13542712872673600000	-0.1354271287	음	0.1354271287	101.5703465451	102	1100101	0000_0000_0110_0101	0_0110_0101	
n+3	-0.00000000000000002498	0.0000000000	0	0.0000000000	0.0000000000	(0)	0	0000_0000_0000_0000	0_0000_0000	Symbol
n+4	0.06420091371232200000	0.0642009137	양	0.0642009137	48.1506852842	48	110000	0000_0000_0011_0000	0_0011_0000	
n+5	-0.04932627602752730000	-0.0493262760	음	0.0493262760	36.9947070206	37	100100	0000_0000_0010_0100	0_0010_0100	
n+6	-0.00000000000000002212	0.0000000000	0	0.0000000000	0.0000000000	(0)	0	0000_0000_0000_0000	0_0000_0000	Symbol
n+7	0.03156805776840270000	0.0315680578	양	0.0315680578	23.6760433263	24	10111	0000_0000_0001_0111	0_0001_0111	
n+8	-0.02574775098543550000	-0.0257477510	음	0.0257477510	19.3108132391	19	10011	0000_0000_0001_0011	0_0001_0011	
n+9	-0.00000000000000001790	0.0000000000	0	0.0000000000	0.0000000000	(0)	0	0000_0000_0000_0000	0_0000_0000	Symbol
n+10	0.01730414582010220000	0.0173041458	양	0.0173041458	12.9781093651	13	1100	0000_0000_0000_1100	0_0000_1100	
n+11	-0.01415300566769760000	-0.0141530057	음	0.0141530057	10.6147542508	11	1010	0000_0000_0000_1010	0_0000_1010	
n+12	-0.00000000000000001302	0.0000000000	0	0.0000000000	0.0000000000	(0)	0	0000_0000_0000_0000	0_0000_0000	Symbol
n+13	0.00928151419730391000	0.0092815142	양	0.0092815142	6.9611356480	7	110	0000_0000_0000_0110	0_0000_0110	
n+14	-0.00739666096745547000	-0.0073966610	음	0.0073966610	5.5474957256	6	101	0000_0000_0000_0101	0_0000_0101	
n+15	-0.00000000000000000821	0.0000000000	0	0.0000000000	0.0000000000	(0)	0	0000_0000_0000_0000	0_0000_0000	Symbol
n+16	0.00446241755819344000	0.0044624176	양	0.0044624176	3.3468131686	3	11	0000_0000_0000_0011	0_0000_0011	

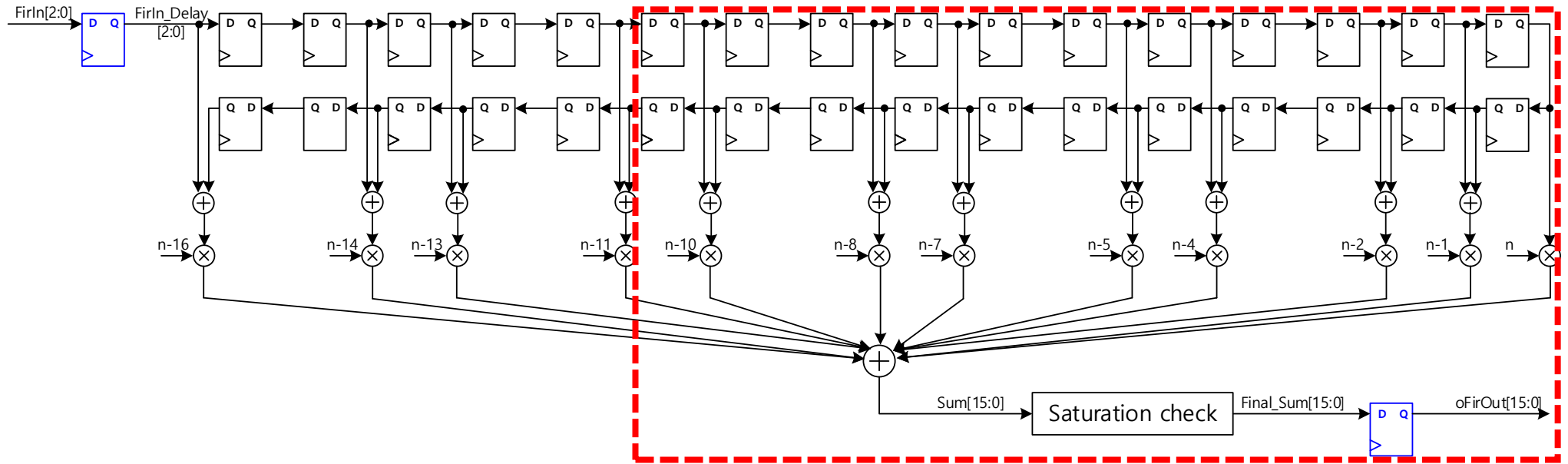
2. FIR Filter algorithm design (팀 Project)

1) HW block diagram

→ Conceptual block diagram



→ 개인 project 구현 구조

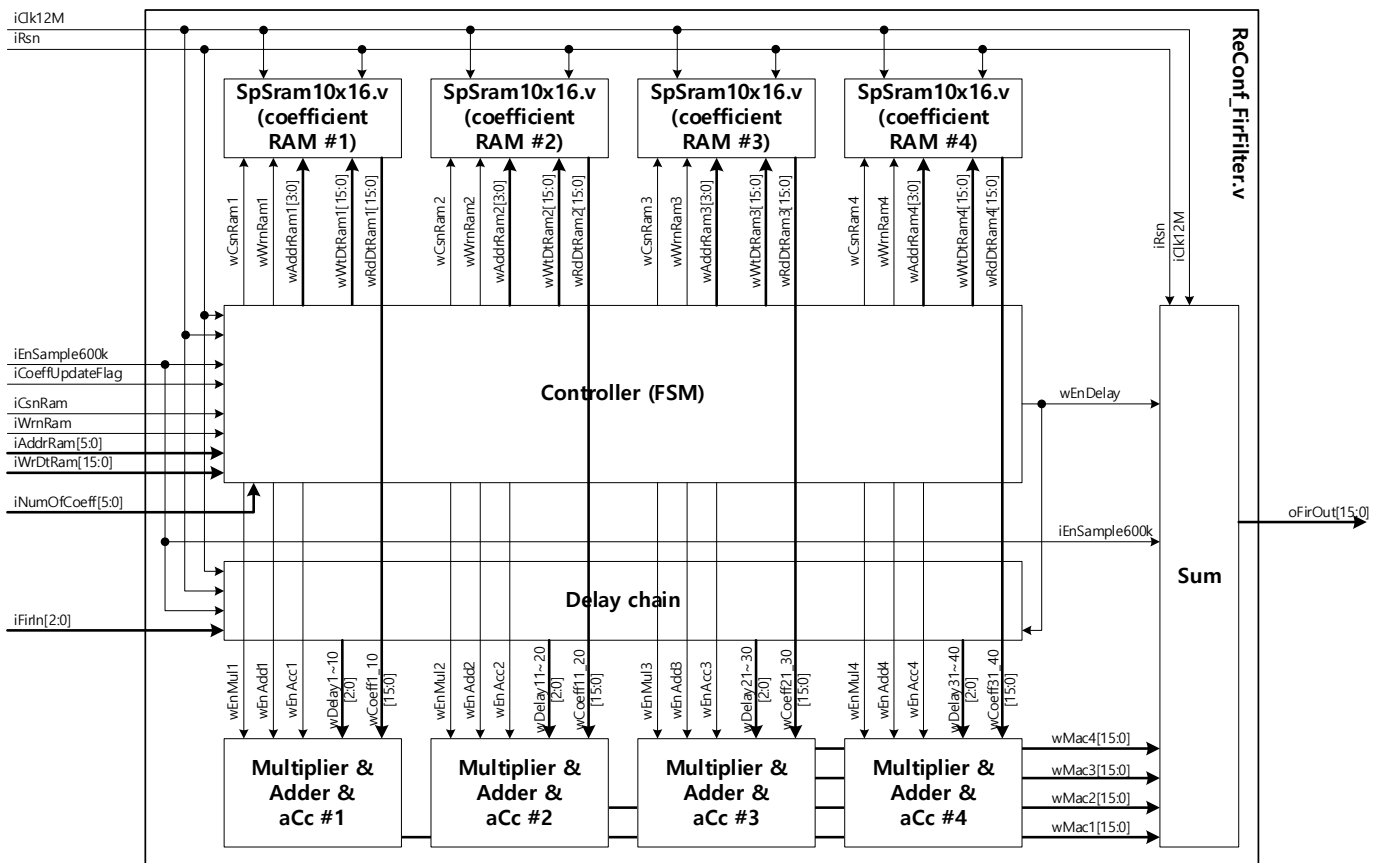


2) Each enable signals

- ➔ Clock input: 12MHz
- ➔ Sampling rate: 600kHz <- $1/20 \times \text{Clock rate}$
- ➔ iFirIn[2:0] pattern: 600kHz/64ea <- $1/1,280 \times \text{Clock rate}$ impulse (Impulse response)
 - iEnSample_600kHz 기준으로 64번에 한번 3'b001 or 3'b111 입력 -> 나머지 63번은 3'b000 입력
- ➔ oFirOut[15:0] pattern:
 - 3'b001 impulse response 예상 값: FIR filter의 coefficient value 출력!!!
 - 3'b111 impulse response 예측 값: FIR filter의 coefficient inversion value 출력!!!

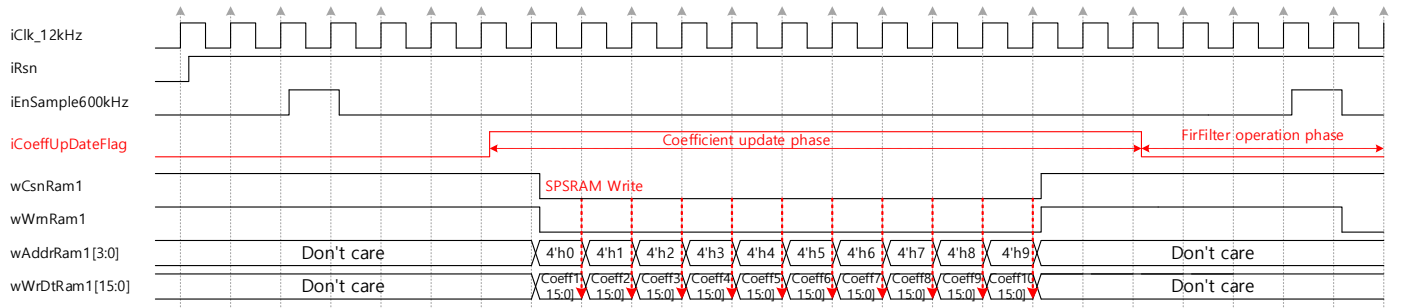
3) Block diagram (Team Project 기본 구조)

- Block diagram update

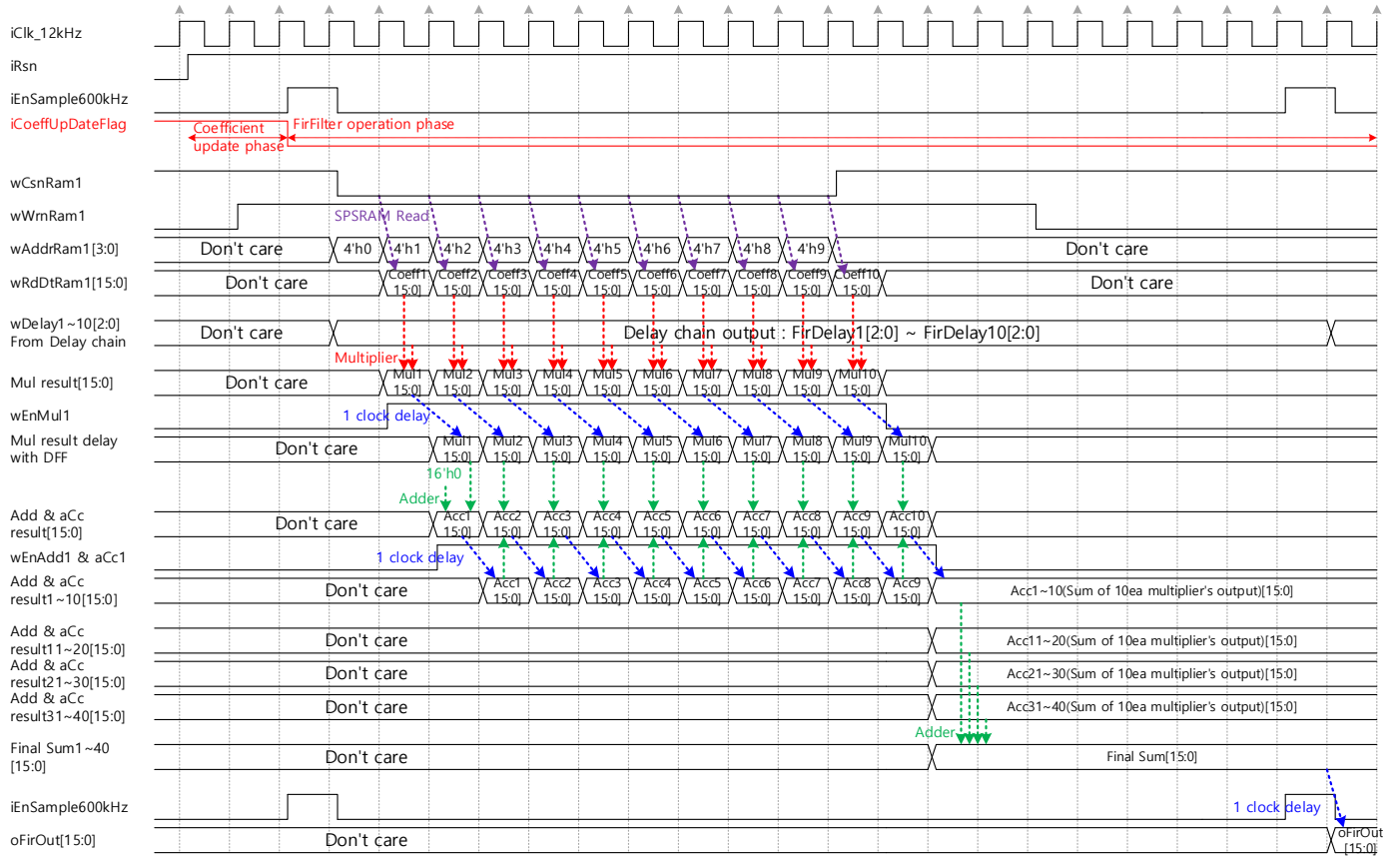


4) Timing diagram

A. Coefficient update phase

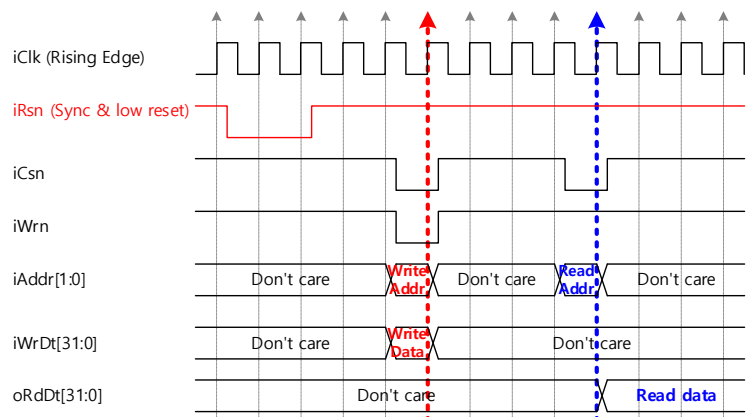
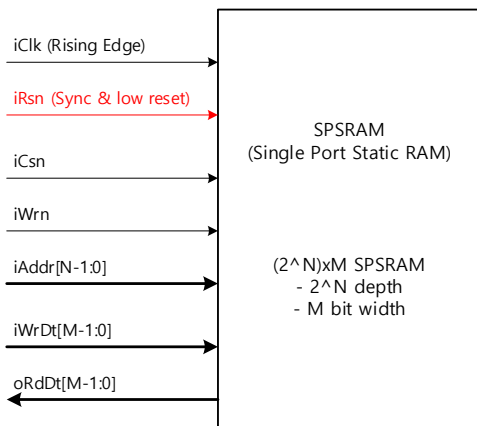


B. Filter operation phase

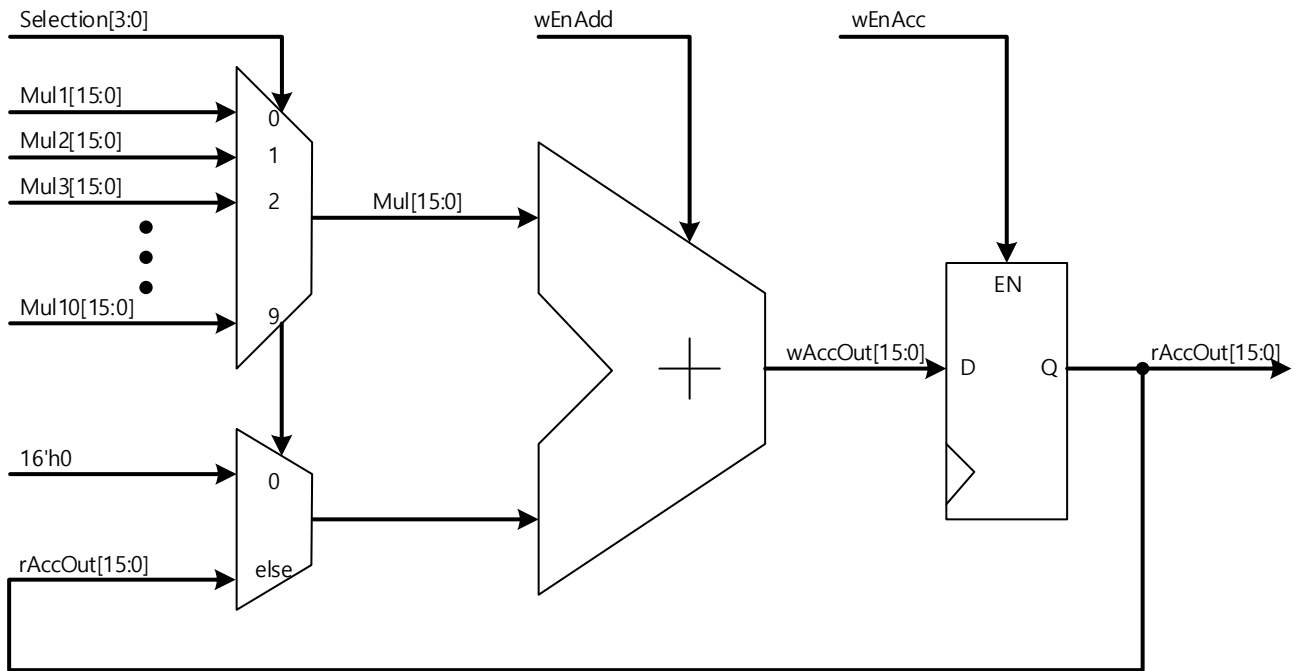


5) SRAM interface & timing

A. SpSram.v 및 SpSram_tb.v 제공 예정



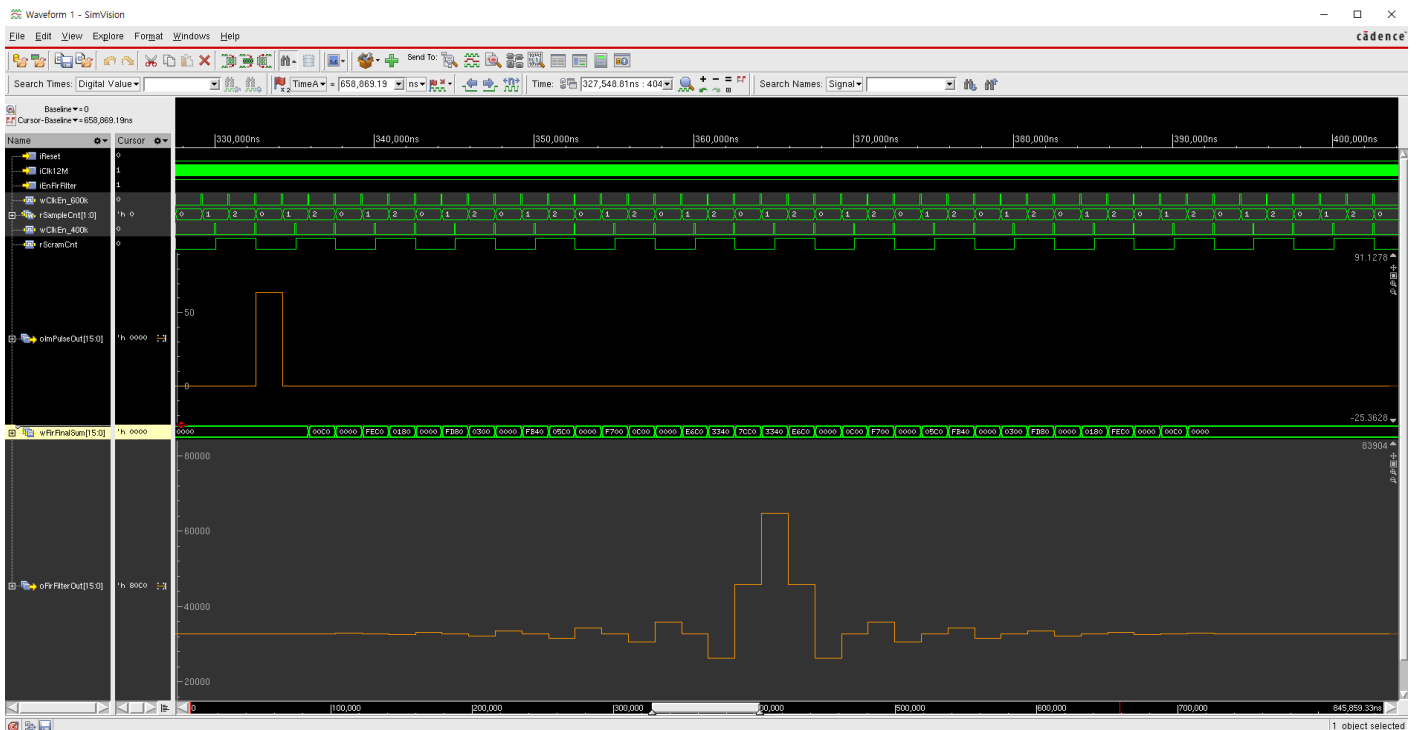
6) Accumulator concept



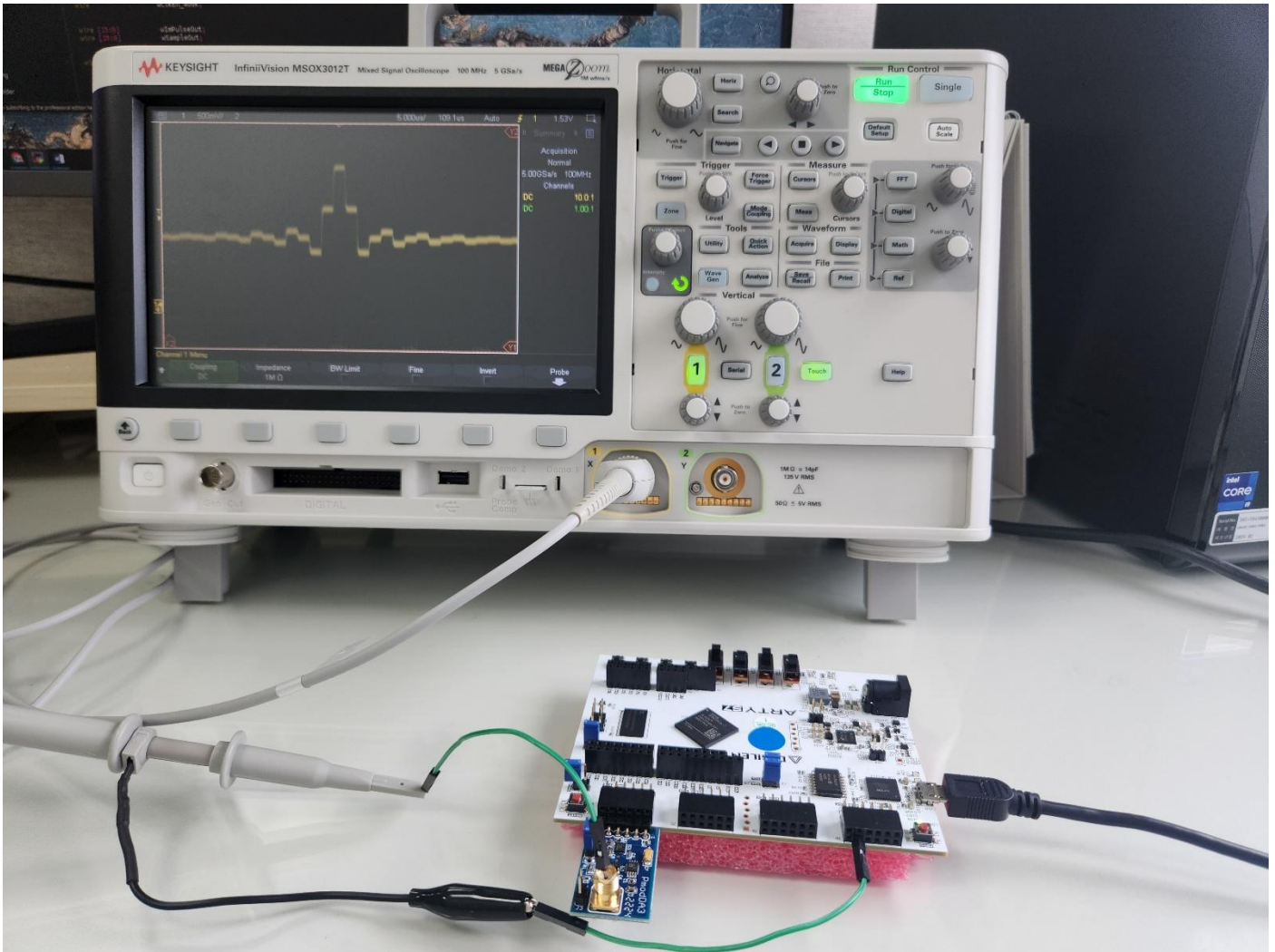
3. FIR Filter result summary (참고용)

1) Impulse response (3'b001)

→ Waveform @simvision

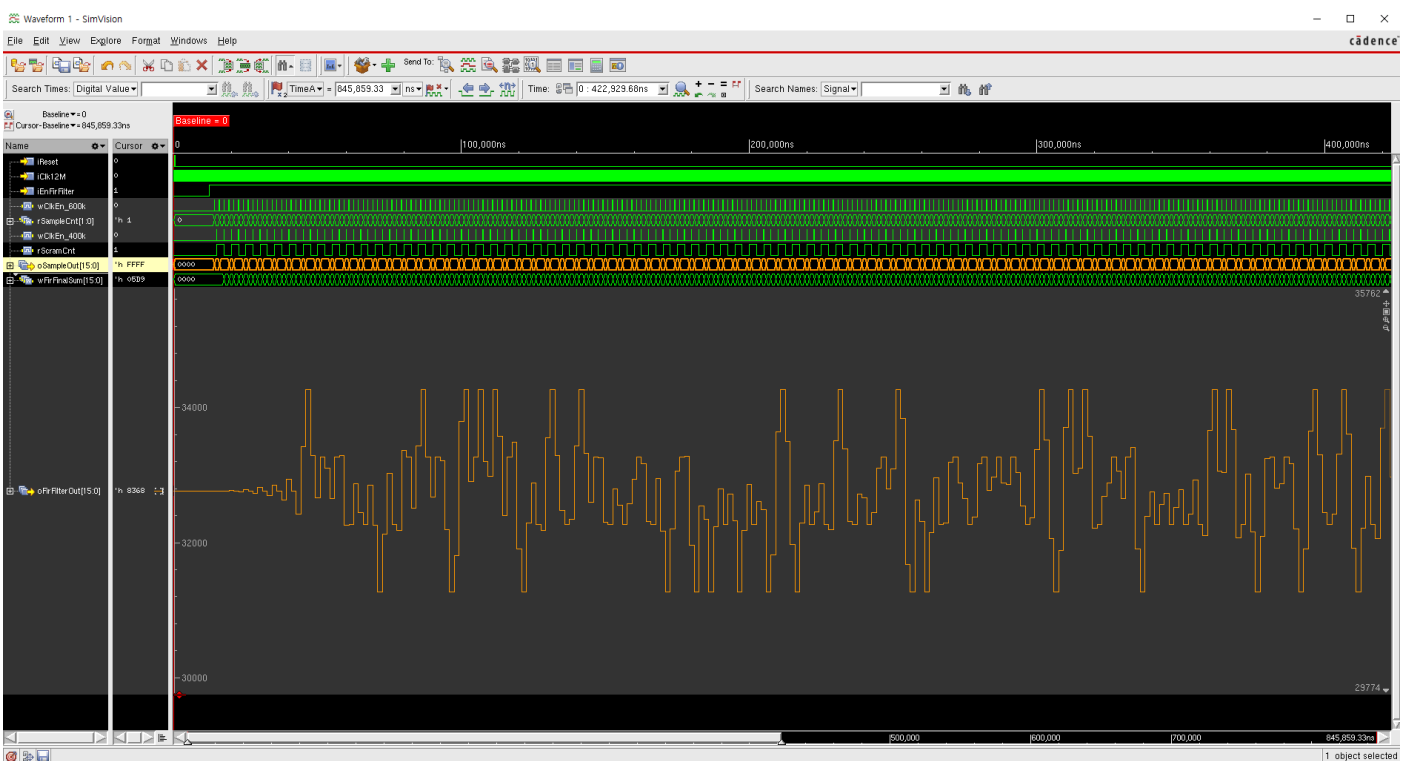


→ Waveform @Oscilloscope

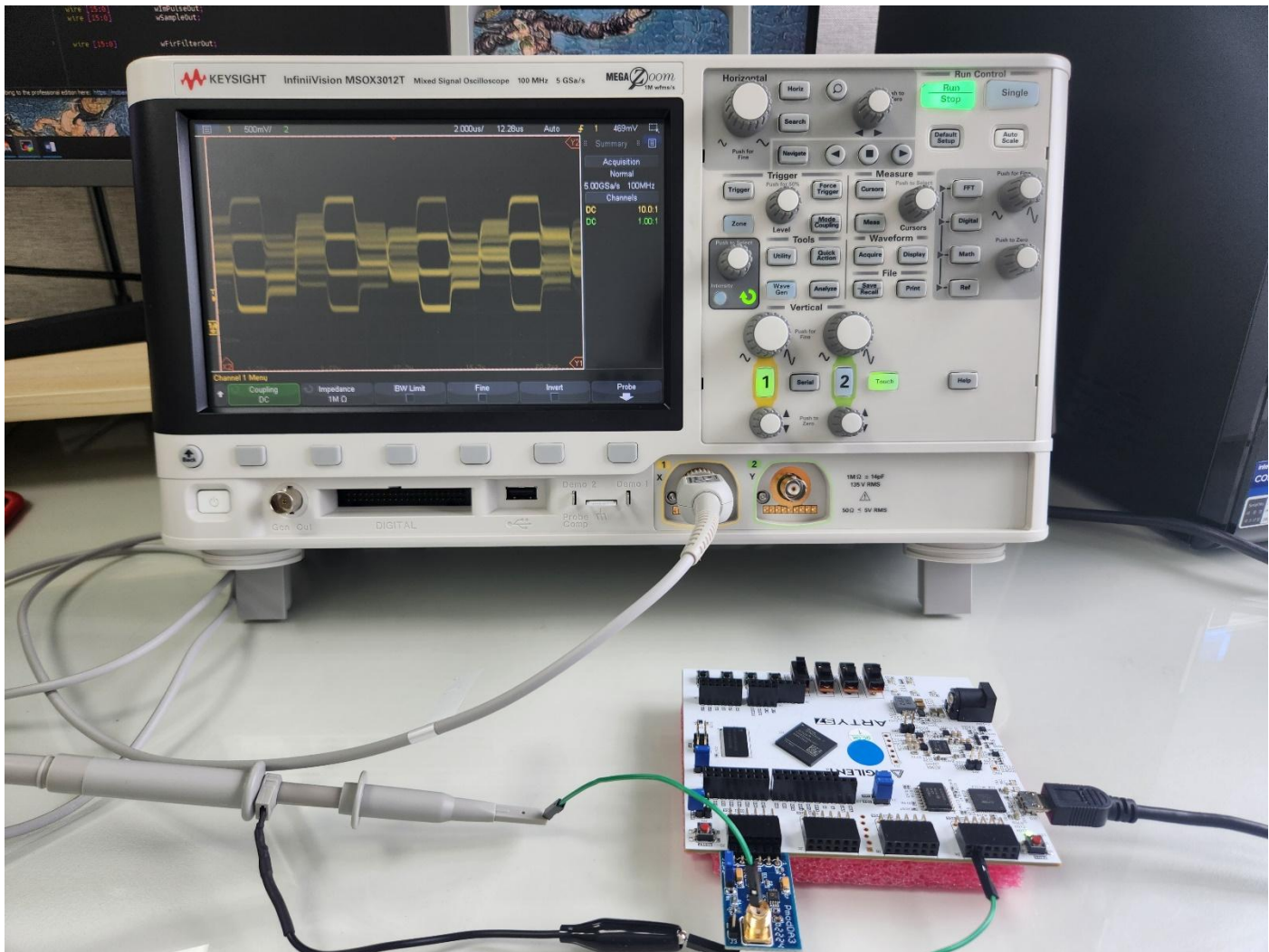


2) Eye diagram

→ Waveform @simvision



➔ Waveform @Oscilloscope



3) Spectrum analysis

➔ TBD.

4. Revision history

Revision History

Date	Version	Description	Author
2024/06/20	V0.0	Initial release	Inchul.Song
2024/11/14	V0.1	Block diagram errors fix	Inchul.Song
2025/11/18	V0.11	2025 project	Inchul.Song

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